



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Reinforcement Learning (AI3000)

Lecture 00: Course Logistics

Karthik P. N.

Assistant Professor, Department of AI

Email: pnkarthik@ai.iith.ac.in

28 July 2026

Timeline

Tue	Fri	Sat	Quiz / TA Session
28 Jul	31 Jul	01 Aug	T0
04 Aug	07 Aug	08 Aug	T1
11 Aug	14 Aug	15 Aug	-
18 Aug	21 Aug	22 Aug	Q1
25 Aug	28 Aug	29 Aug	T2
01 Sep	04 Sep	05 Sep	Q2
08 Sep	11 Sep	12 Sep	T3
15 Sep	18 Sep	19 Sep	T4
22 Sep	25 Sep	26 Sep	Q3
29 Sep	02 Oct	03 Oct	T5
06 Oct	09 Oct	10 Oct	Q4
13 Oct	16 Oct	17 Oct	-
20 Oct	23 Oct	24 Oct	-
27 Oct	30 Oct	31 Oct	T6
03 Nov	06 Nov	07 Nov	Q5
10 Nov	13 Nov	14 Nov	-

T0-T6: TA Sessions

Q1-Q5: Quizzes

No session / holiday



Mid-Sem Exam: 16 Sep



Final Exam: 18 Nov

Course at a Glance

Course	Reinforcement Learning (AI3000)
Credits	3
Focus	Learning to make good sequential decisions from interaction with an environment
Foundations	Probability, linear algebra, optimization, and foundations of machine learning
Components	Mathematical foundations, algorithms, and implementation exercises

🎯 Learning Objective

By the end of the course, you should be able to model sequential decision-making problems and derive, analyse, and implement core reinforcement-learning algorithms

Assessment	Weightage
Quizzes	40%
Best 4 of 5 scores will be considered	
Mid-Sem Exam	30%
Final Exam	30%

Student Brigade (Teaching Assistants)



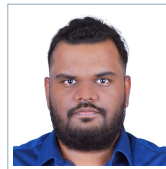
Mahvith Akshintala



**Deepshikha
Urmaliya**



Prantik Pramanick



**Vedant Prakash
Shenoy**



U Adithyan

 **Course Website:** karthikpn.com/#/teaching/Reinforcement-Learning-2026

Outline

Foundations and Bandits

- Agent–environment interaction
- Rewards, returns, and policies
- Exploration versus exploitation
- Multi-armed bandits

Markov Decision Processes

- Value functions and Bellman equations
- Policy evaluation and improvement
- Value and policy iteration

Model-Free Learning

- Monte Carlo methods
- Temporal-difference learning
- SARSA and Q-learning

Approximation and Deep RL

- Function approximation
- Policy-gradient methods
- Actor–critic methods
- Deep reinforcement learning